Vehicle Insurance Customer Lifetime Value Analysis

BUSINESS OBJECTIVE:
To identify which policy and vehicle characteristics are associated with higher Customer Lifetime Value, enabling the insurer to prioritize pricing, coverage, and product strategies that grow value from its highest-potential segments.

DATASET:

- Source: Kaggle UCI ML Repository
- Records: 9,134 customers
- Variables Analysed: 8 (6 categorical, 2 numerical)
- Target Variable: Customer Lifetime Value (CLV; The estimated total value a customer is expected to generate for the insurer over the course of their relationship with the company. *This has been provided as a pre-calculated metric in the source dataset.)*

ANALYSIS APPROACH:

- Initial data inspection in Python
- Variable selection based on business relevance — focusing on policy and vehicle characteristics that represent actionable levers for the insurance company
- Numerical variable selection further guided by correlation analysis with CLV
- All visualization and segmentation done in Power BI across 4 pages — Overview, Policy Segmentation, Channel & Policy Volume, and Numerical Analysis

DASHBOARD STRUCTURE:

Page 1 — Overview: High-level portfolio summary including Average Customer Lifetime Value, total customer count, average monthly premium, CLV distribution histogram, and CLV by vehicle class

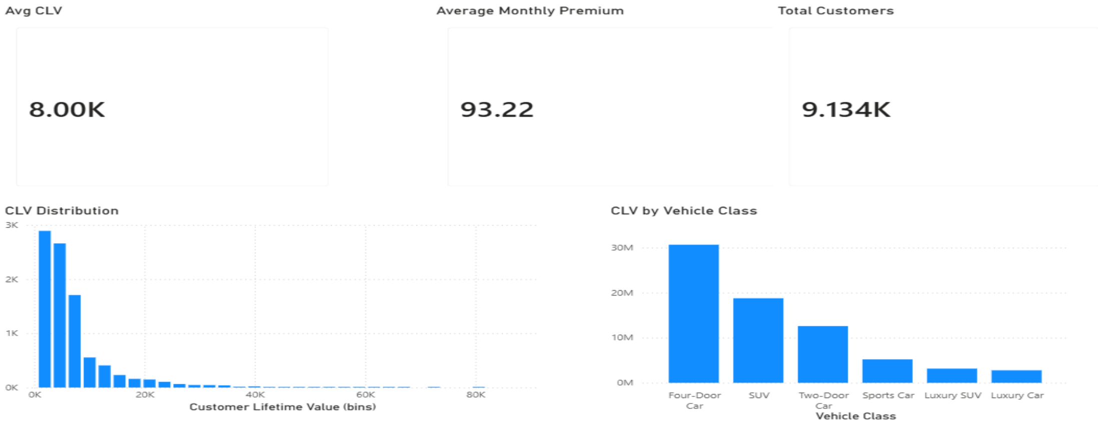
Page 2 — Policy Segmentation: CLV and customer count analysis by vehicle class, policy type, and coverage level

Page 3 — Channel and Policy Volume: CLV and customer count analysis by renewal offer type, sales channel, and number of policies

Page 4 — Numerical Analysis: Scatter plot of Monthly Premium Auto versus Customer Lifetime Value illustrating the positive association between the two variables"

KEY INSIGHTS:

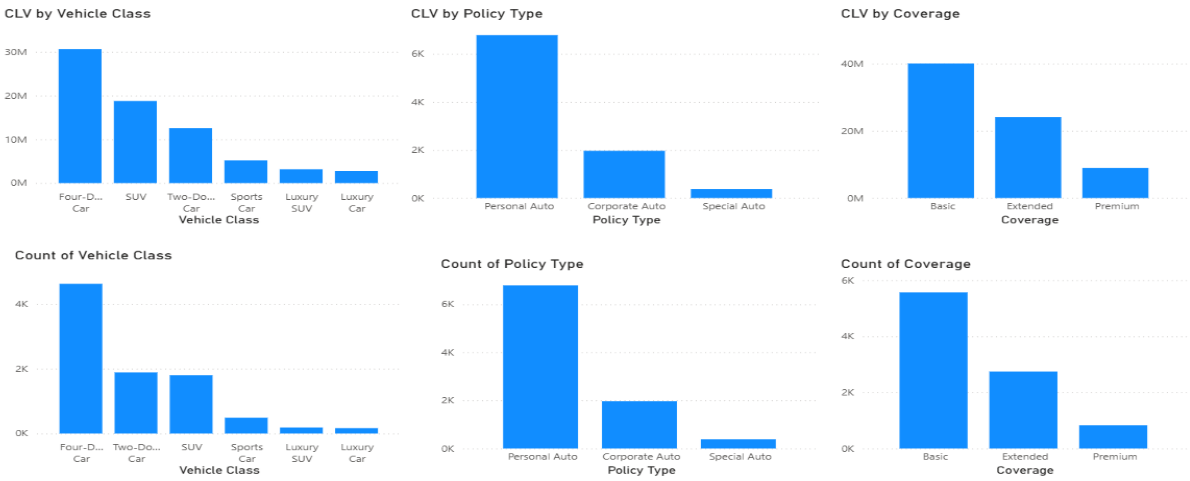
- Overview



  - ➢ CLV distribution is right-skewed with average CLV of 8,000 across 9,134 customers

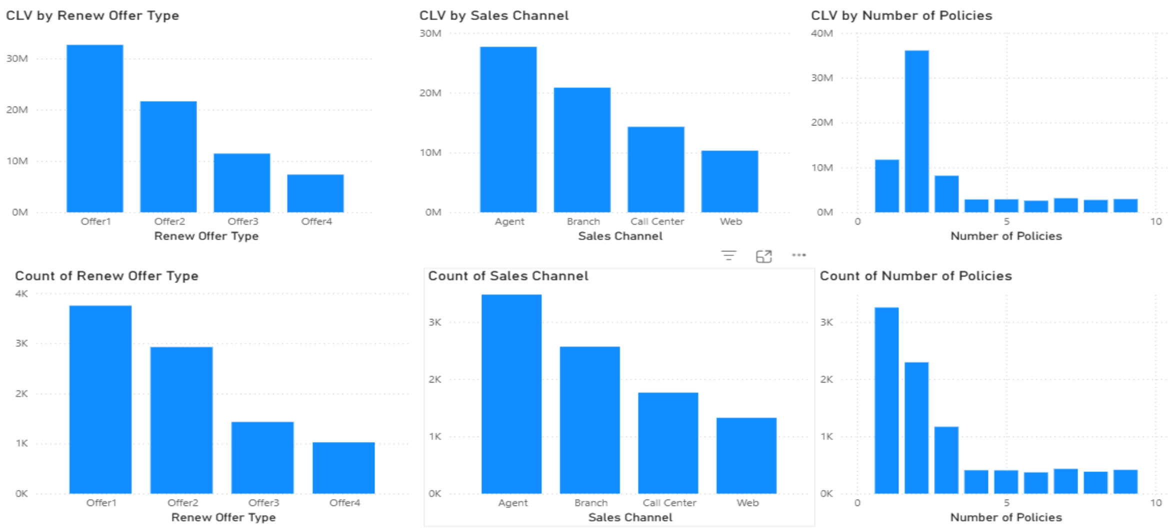
<u>Note</u>: *The dataset contains U.S. states (e.g., California, Oregon, Washington). Therefore, Customer Lifetime Value figures and other currency dependent features are assumed to be denominated in U.S. dollars (USD), although the dataset documentation does not explicitly specify the currency.*

- Policy Segmentation



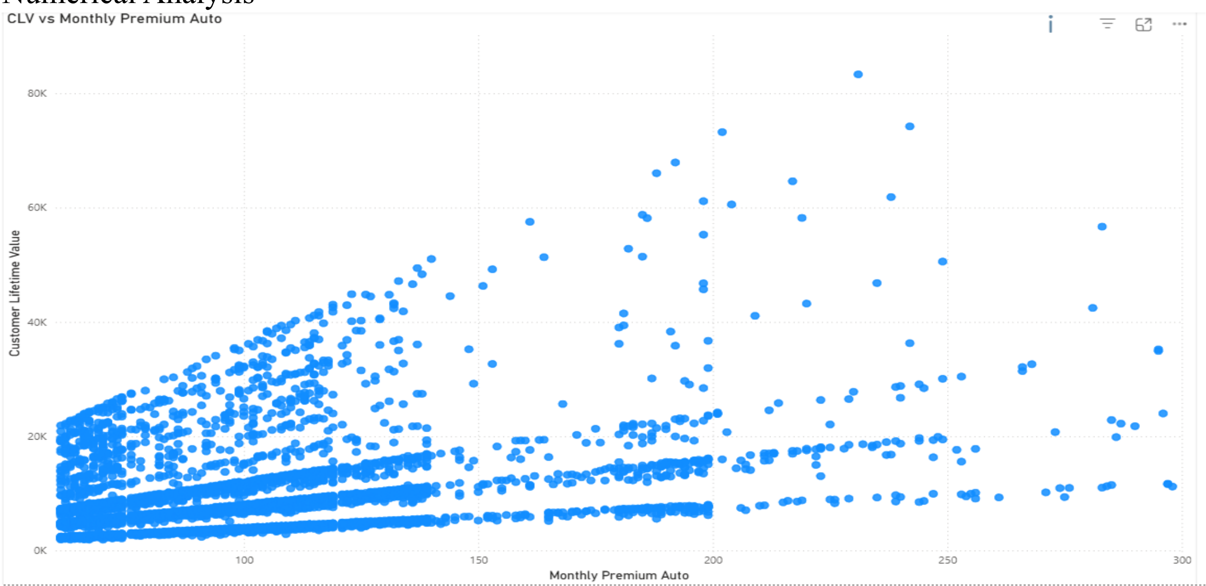
  - ➢ Personal Auto policies and Basic Coverage dominate portfolio volume and value
  - ➢ Four-Door Car segment contributes highest total CLV driven by volume; SUV customers despite being third in count rank second in total CLV — indicating higher individual customer value and making them a priority retention segment

- Channel and Policy Volume



- ➢ Agent sales channel leads all channels in both customer volume and total CLV generation — indicating it is the most effective channel for both reaching and retaining high value customers
- ➢ Customers holding 2 policies contribute the highest total CLV despite lower count compared to single policy holders — suggesting cross-selling a second policy is a high value strategy for the business

- Numerical Analysis



Monthly Premium Auto shows a positive association with CLV. The scatter plot exhibits a fan-shaped pattern, indicating heteroscedasticity, where the variability of CLV increases at higher premium levels.

*Heteroscedasticity: A condition where the variability (spread) of a variable is not constant across the range of another variable.

- **Conclusion**

  This analysis explored Customer Lifetime Value across key policy and vehicle dimensions in a vehicle insurance portfolio. It revealed that Personal Auto policies, Basic Coverage, Agent sales channel and Four-Door Car segment dominate both customer volume and total CLV. SUV customers and customers with 2 policies demonstrated disproportionately higher value relative to their count — highlighting cross-selling and targeted retention as potential business opportunities.

  The analysis also indicated a positive association between Monthly Premium Auto and CLV. The scatter plot exhibited a fan-shaped pattern, suggesting heteroscedasticity, where the variability of CLV increases at higher premium levels.

  Overall, the analysis provides a data driven foundation for customer segmentation in vehicle insurance, enabling the business to focus on high value policy and vehicle characteristics that are within its control.